



ReadyNest

Summer Internship Project
Report

PATNA BUSINESS ANALYTICS & PREDICTIVE ANALYTICS USING PYTHON AND POWER BI

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DATASET DESCRIPTION

The dataset used in this project consists of information collected from various local businesses located in Patna, Bihar. The objective of the dataset is to evaluate business performance using customer feedback, digital presence, and operational characteristics.

The dataset contains **49 business records** representing multiple business categories including restaurants, cafés, salons, gyms, medical stores, and retail shops. Each record contains important attributes that describe the business and its online presence.

The data was organized in Microsoft Excel before being imported into Python for preprocessing and analysis.

TECHNOLOGY STACK

The project integrates multiple technologies to complete the analytics workflow.

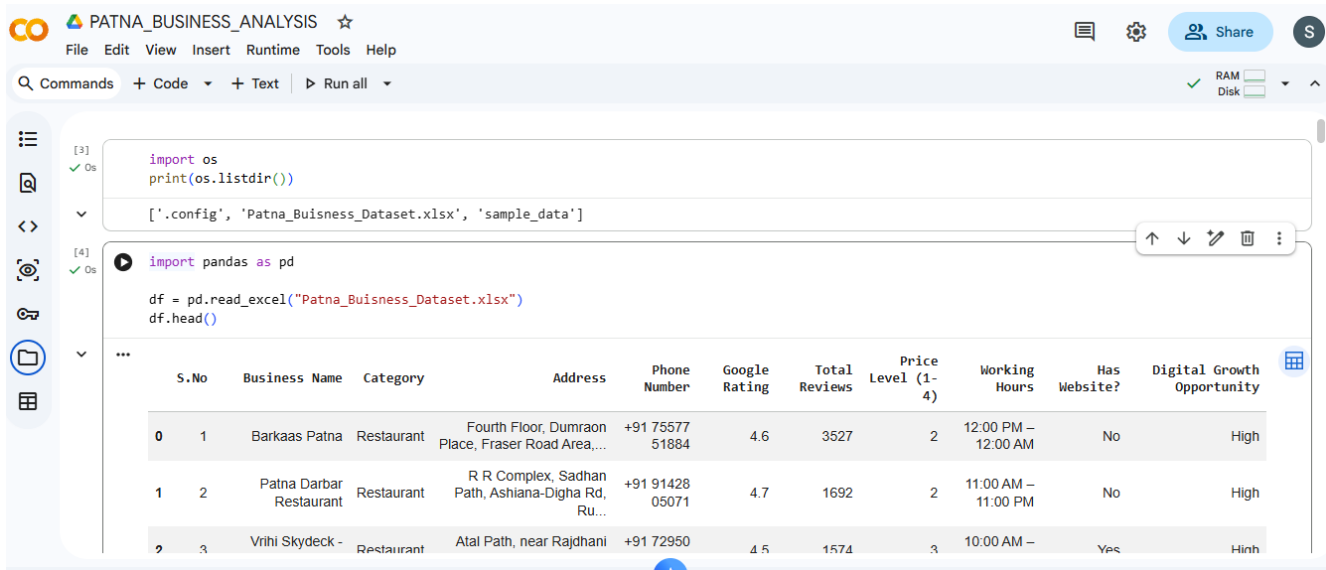
- **Python** – Data preprocessing, visualization, machine learning.
- **Google Colab** – Cloud-based development environment.
- **Pandas** – Data manipulation and preprocessing.
- **NumPy** – Numerical computation.
- **Matplotlib** – Data visualization.
- **Seaborn** – Statistical visualization.
- **Scikit-learn** – Machine learning.
- **Microsoft Power BI** – Interactive dashboard creation.

DATA CLEANING & PREPROCESSING

Before model training, the dataset underwent preprocessing to improve quality and consistency.

The following steps were performed:

- Imported the dataset into Google Colab.
- Checked data types and column names.
- Verified missing values.
- Checked duplicate records.
- Encoded categorical variables using Label Encoding.
- Selected features for machine learning.
- Split the dataset into training and testing sets.



The screenshot shows the Google Colab interface for a notebook titled 'PATNA_BUSINESS_ANALYSIS'. The code editor displays two code cells. Cell [3] imports the 'os' module and prints the directory listing. Cell [4] imports 'pandas as pd', reads the 'Patna_Business_Dataset.xlsx' file, and displays the first few rows of the dataset using 'df.head()'. Below the code, a table preview of the dataset is shown with 12 columns: S.No, Business Name, Category, Address, Phone Number, Google Rating, Total Reviews, Price Level (1-4), Working Hours, Has Website?, and Digital Growth Opportunity. The first three rows of data are visible.

S.No	Business Name	Category	Address	Phone Number	Google Rating	Total Reviews	Price Level (1-4)	Working Hours	Has Website?	Digital Growth Opportunity
0	1	Barkaas Patna Restaurant	Fourth Floor, Dumraon Place, Fraser Road Area,...	+91 75577 51884	4.6	3527	2	12:00 PM – 12:00 AM	No	High
1	2	Patna Darbar Restaurant	R R Complex, Sadhan Path, Ashiana-Digha Rd, Ru...	+91 91428 05071	4.7	1692	2	11:00 AM – 11:00 PM	No	High
2	3	Vrihi Skydeck - Restaurant	Atal Path, near Rajdhani	+91 72950	4.5	1574	3	10:00 AM –	Yes	High

EXPLORATORY DATA ANALYSIS

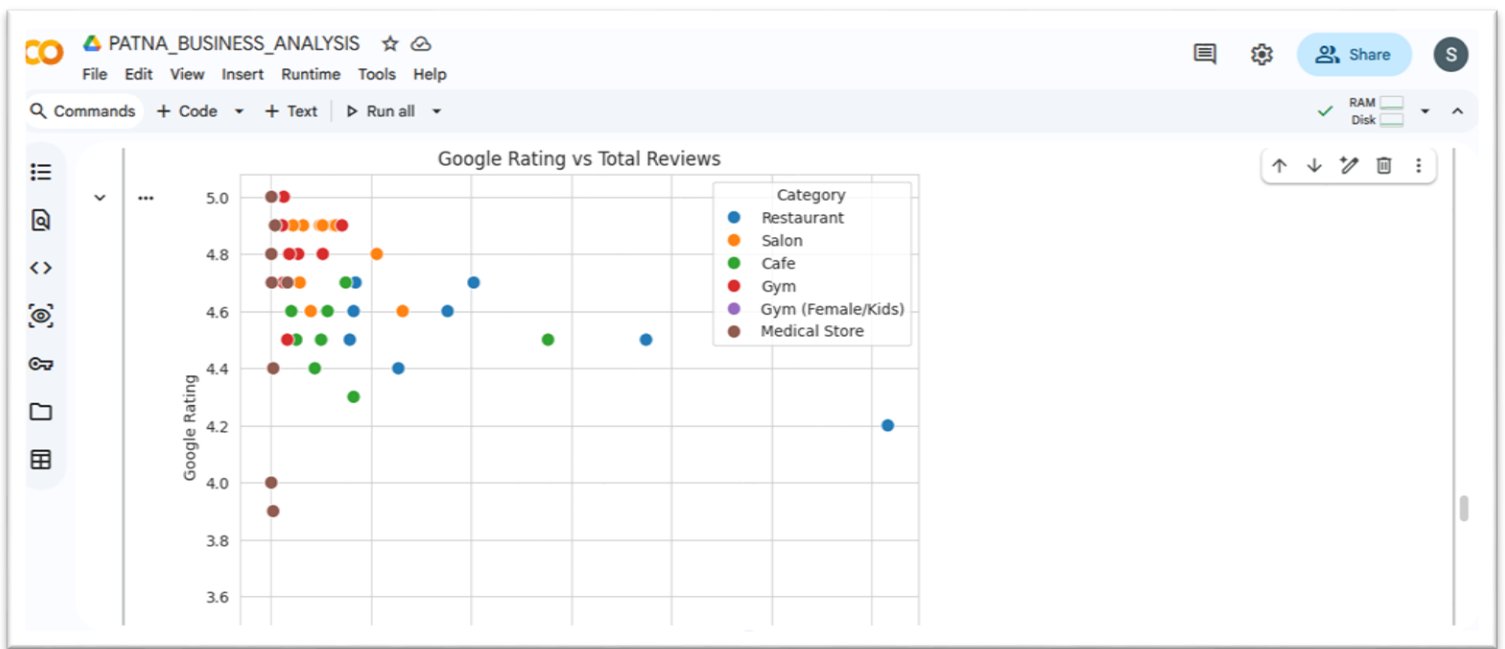
Exploratory Data Analysis (EDA) was performed to understand relationships within the dataset before building the predictive model.

Rating Distribution

Most businesses received ratings above 4.5, indicating high customer satisfaction.

Scatter Plot

The scatter plot showed a positive relationship between Google Rating and Total Reviews. Businesses with more reviews generally maintained strong ratings.



Correlation Heatmap

The heatmap identified relationships between numerical variables, helping in feature selection.

Website Analysis

Several highly rated businesses did not have official websites, indicating opportunities for digital expansion.

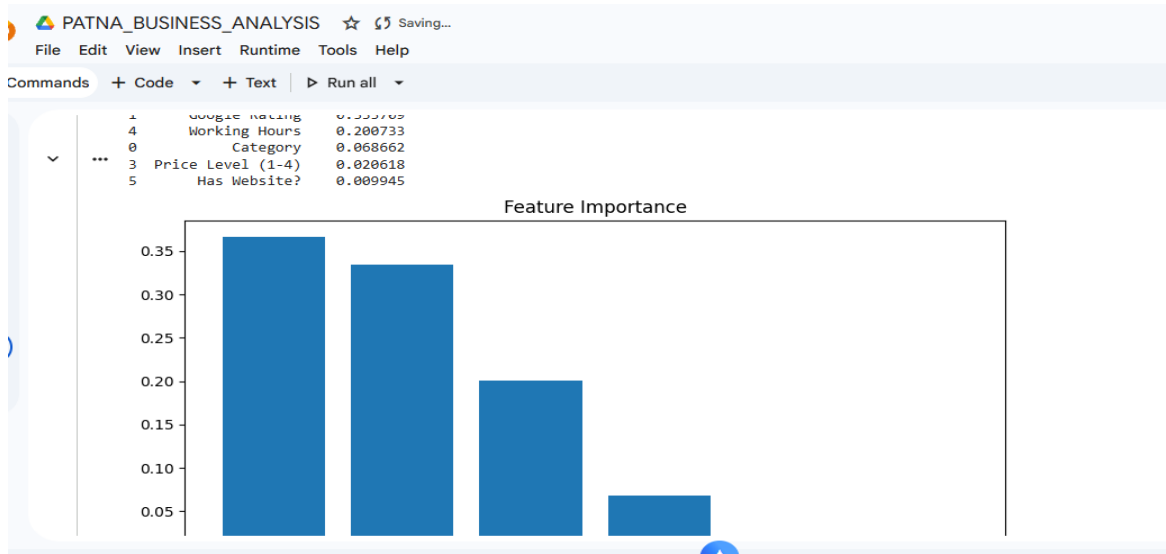
FEATURE IMPORTANCE ANALYSIS

Feature Importance identified the variables that contributed most to the model's predictions.

The analysis indicated that:

- Working Hours
- Business Category
- Google Rating
- Total Reviews
- Website Availability

were among the most influential features.

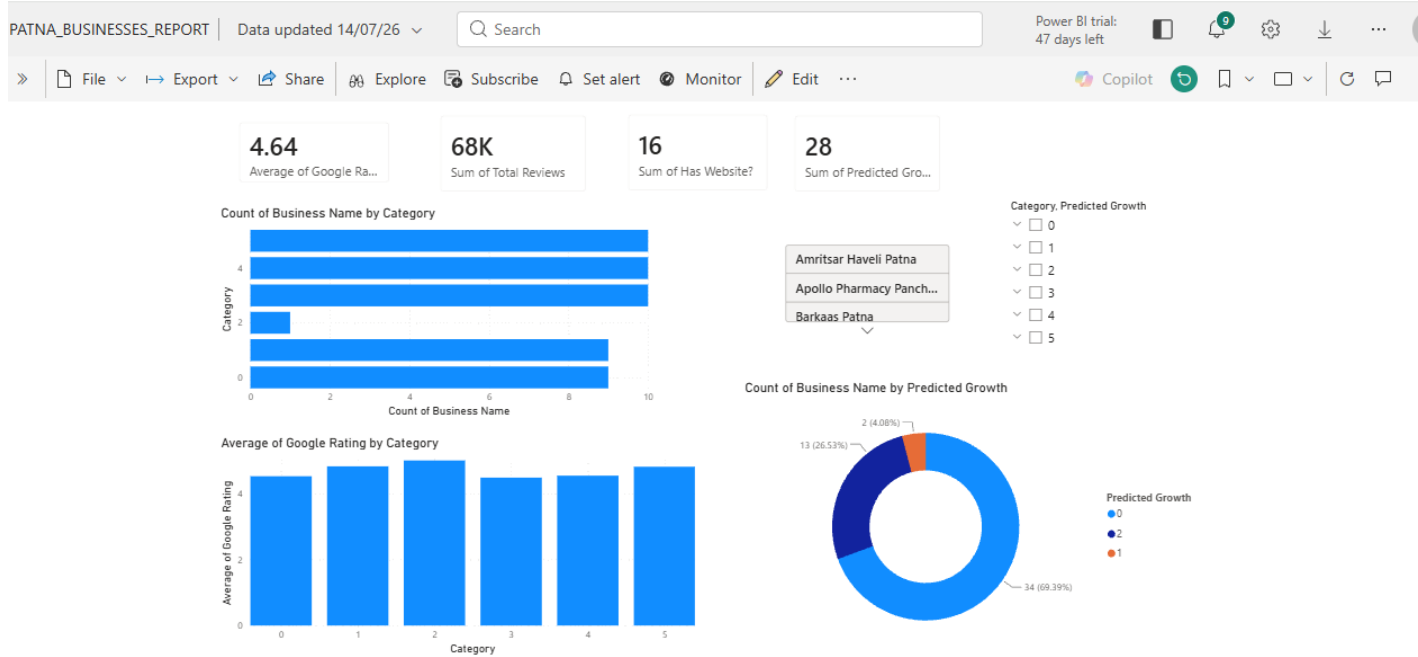


POWER BI DASHBOARD

Three dashboard pages were created.

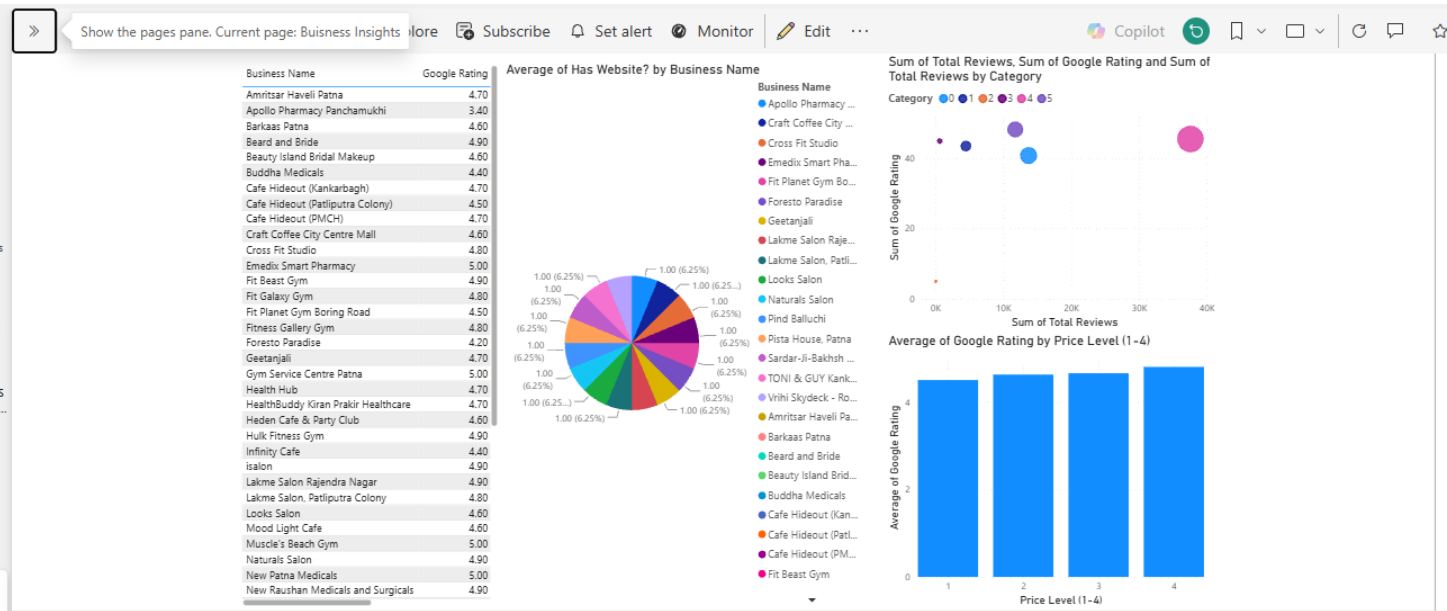
Executive Dashboard

Displays KPIs including Average Rating, Total Reviews, Businesses with Website, and Growth Opportunity.



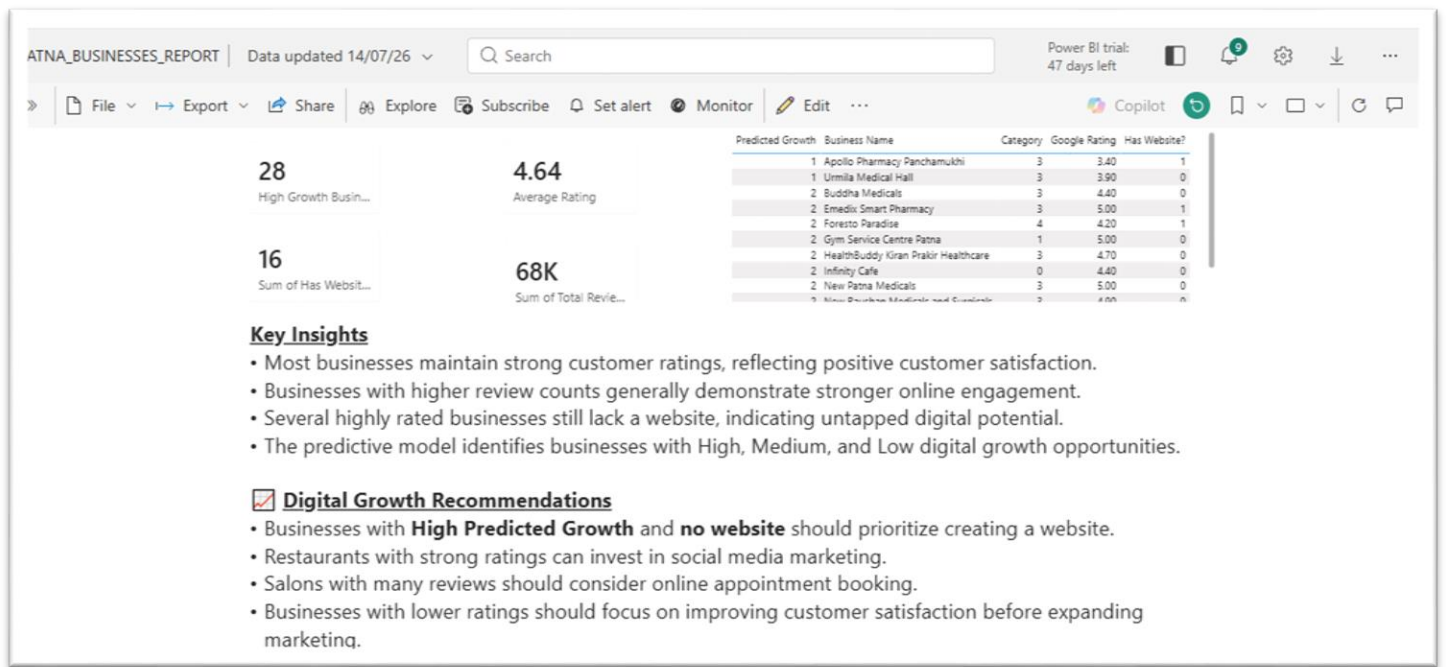
Business Insights Dashboard

Includes scatter plots, business comparison tables, and website analysis.



Predictive Analytics Dashboard

Displays prediction summaries, recommendation tables, and interactive filters.



ADVANCED DAX MEASURES

The following DAX measures were created:

- Average Rating
- Total Reviews

- Website Count
- High Growth Businesses
- Business Rank
- Category Performance

These measures enabled dynamic filtering and interactive analysis.

BUSINESS INSIGHTS

Key insights obtained from the analysis include:

- Most businesses achieved ratings above 4.5.
- Businesses with more customer reviews generally maintained stronger ratings.
- Several businesses lacked websites despite having excellent customer ratings.
- Restaurants and cafés represented the largest business category.
- Predictive analytics identified businesses with strong digital growth opportunities

RECOMMENDATIONS

Restaurants

- Improve online ordering systems.
- Encourage customer reviews.

Salons

- Introduce online appointment booking.
- Increase social media marketing.

Cafés

- Enhance digital advertising.
- Develop customer loyalty programs.

Medical Stores

- Provide medicine availability online.
- Improve digital customer support.

General Recommendations

- Develop official websites.
- Improve Google Business profiles.

- Encourage customer feedback.
- Invest in digital marketing.

FUTURE SCOPE

The project can be extended by:

- Integrating Google Maps API
- Real-time dashboard refresh
- Sentiment Analysis using NLP
- Sales Forecasting
- Customer Segmentation
- Deep Learning Models

CONCLUSION

This project successfully demonstrated the integration of Business Analytics, Machine Learning, and Business Intelligence using Python and Microsoft Power BI.

The workflow included data collection, preprocessing, exploratory analysis, predictive modeling using a Random Forest Classifier, and dashboard development. The Power BI dashboards transformed the analytical results into interactive visualizations that support decision-making.

Although the dataset contains only 49 businesses, the project demonstrates a complete end-to-end analytics pipeline that can be scaled to larger datasets. The project strengthened practical skills in Python, machine learning, data visualization, DAX, and dashboard development while highlighting the importance of data-driven business decisions.

Thank You